Revision chemistry — Answers

Grade 7 · IB MYP 2 Sciences · Unit 1 Chemistry's Secrets Unlocked · answers to the study guide

How to use the answers

- These are model answers at Grade 7 level. Other answers that use correct chemistry and the right key words are also right.
- Green notes (*A strong answer includes …*) show what a top answer has.
- Mark your study guide and fill in the **Self-check** column: what you got right and what you need to fix.

A · Atoms, elements, compounds and mixtures

#	Question	Model answer
A1	What is an atom?	An atom is the smallest particle of an element. Everything is made of atoms.
A2	Element + two examples with symbols	A pure substance made of only one type of atom. It cannot be broken down into simpler substances. Examples: oxygen (O), iron (Fe), carbon (C), gold (Au), sodium (Na).
A3	Compound + two examples with formulas	A substance made of two or more different elements chemically joined (bonded) in a fixed ratio. Examples: water (H_2O), carbon dioxide (CO_2), sodium chloride (NaCl).
A4	Mixture + two everyday examples	Two or more substances mixed together but not chemically joined. Examples: air, salt water, sand and water, orange juice, soil.
A5	Two differences: compound vs mixture	**1** A compound has a fixed ratio of elements (water is always H_2O); a mixture can have any amounts. **2** A compound can only be separated by a chemical reaction; a mixture can be separated by physical methods (filtering, evaporating, a magnet, distillation). (Also accept: a compound has new properties; a mixture keeps the properties of its parts.) *A strong answer includes: two clear comparisons, each saying what is true for a compound AND for a mixture.*
A6	Sort: element, compound or mixture	**Elements:** oxygen (O_2), iron (Fe) **Compounds:** water (H_2O), sodium chloride (NaCl), carbon dioxide (CO_2) **Mixtures:** air, salt water, orange juice
A7	Iron + sulfur → iron sulfide	A compound has different properties from the elements it is made from. Iron sulfide is not grey or magnetic like iron and not yellow like sulfur, because the atoms are chemically joined to make a new substance. *A strong answer includes: "new substance" / "different properties" + evidence from the colours and the magnetism.*
A8	24 carat ring vs 9 carat ring	The 24 carat ring is an element: it contains only gold atoms. The 9 carat ring is a mixture (an alloy): gold, copper and silver are mixed but not chemically joined.

#	Question	Model answer
A9	Dry salt back from sand, salt and water	**1** Stir so all the salt dissolves in the water (dissolving). **2** Filter: the sand stays in the filter paper (residue); the salt water passes through (filtrate). **3** Evaporate the water by heating the filtrate (or leave it in a warm place); salt crystals are left. *A strong answer includes: the correct order and the names filtration and evaporation.*
A10	Box of particles	It is a **mixture**. There are different substances in the box: single atoms (an element), joined identical atoms (molecules of an element) and joined different atoms (a compound). A pure substance would contain only one kind of particle.

B · Acids and alkalis

Property	Acids	Alkalis
pH range	below 7 (0–6)	above 7 (8–14)
Blue litmus paper	turns red	stays blue
Red litmus paper	stays red	turns blue
Taste (never test in the lab!)	sour	bitter
Feel on skin	can sting or burn (strong acids are corrosive)	soapy or slippery; strong alkalis are corrosive
Everyday examples	lemon juice, vinegar, cola, orange juice	soap, toothpaste, baking soda, bleach, oven cleaner
Lab examples	hydrochloric acid, sulfuric acid, nitric acid	sodium hydroxide, potassium hydroxide, calcium hydroxide

#	Question	Model answer
B1	Alkali vs base	An alkali is a base that dissolves in water. A base is any substance that neutralises an acid, so all alkalis are bases, but not all bases dissolve in water (e.g. copper oxide, magnesium hydroxide).
B2	Why the "corrosive" symbol?	Both can attack and destroy living tissue (skin, eyes) and damage materials like metals, so they must be handled with care, gloves and goggles.
B3	"Acids are always dangerous."	Many foods contain weak, dilute acids that are safe to eat, e.g. lemon juice (citric acid), vinegar (ethanoic acid), apples, oranges and yoghurt. Only strong or concentrated acids are dangerous. *A strong answer includes: two food examples + the idea of weak or dilute acids.*
B4	Sodium hydroxide on a hand	Tell the teacher straight away and rinse the hand under plenty of cold running water for several minutes. The water dilutes and washes away the alkali, so it does less damage to the skin. Do not try to neutralise it with acid on the skin.

#	Question	Model answer
B5	Two unlabelled liquids	Put a drop of each on red and blue litmus paper (or add universal indicator). The acid turns blue litmus red (indicator red, orange or yellow). The alkali turns red litmus blue (indicator blue or purple). Wear goggles; never taste or touch.
B6	Strong alkali soap	Strong alkalis are corrosive. They react with the natural oils and proteins in skin, remove the oils and damage the skin, so it feels dry and sore. Modern soaps are only weakly alkaline.

C · The pH scale

pH	0–2	3–6	7	8–11	12–14
Universal indicator colour	red	orange / yellow	green	blue	purple (violet)
Type	strong acid	weak acid	neutral	weak alkali	strong alkali
Example	stomach acid, hydrochloric acid	vinegar, orange juice	pure water	baking soda, soap	oven cleaner, sodium hydroxide

#	Question	Model answer
C1	What does pH measure?	pH measures how acidic or alkaline a solution is. The scale runs from 0 to 14.
C2	pH of pure water	pH 7. A substance with pH 7 is neutral.
C3	Most acidic → most alkaline	stomach acid (1.5) → lemon juice (2) → milk (6.5) → pure water (7) → baking soda solution (9) → oven cleaner (13)
C4	Universal indicator vs litmus	Litmus only shows whether something is an acid or an alkali. Universal indicator shows a range of colours, so it also tells you the pH number, i.e. how strongly acidic or alkaline it is.
C5	Rain at pH 5.6 vs pH 4	The rain at pH 4 is more acidic (lower pH). This acid rain can slowly dissolve (wear away) limestone and marble statues and buildings, and can make lakes too acidic for fish and plants to survive.
C6	Blueberries need pH 4.5–5.5; soil is pH 7	No: the soil is neutral, not acidic enough. The gardener should add a weakly acidic substance (e.g. acidic compost, pine needles or a soil acidifier) to lower the pH into the 4.5–5.5 range.
C7	Why such a strong acid in the stomach?	The acid kills most harmful bacteria in food and helps to break down (digest) food, especially proteins.

D · Neutralisation

#	Question	Model answer
D1	Neutralisation	A reaction between an acid and an alkali (or base) that cancels out their properties and makes a salt and water. The pH moves towards 7.
D2	acid + alkali → ? + ?	acid + alkali → **salt** + **water**

#	Question	Model answer
D3	Adding NaOH to HCl with universal indicator	At the start the solution is red (pH about 1). As the alkali is added the pH rises and the colour changes red → orange → yellow → green. At green the pH is 7 and the solution is neutral. (If too much alkali is added it turns blue, then purple.)
D4	Bee sting vs wasp sting	Bee sting (acidic): use baking soda (weakly alkaline) to neutralise the acid. Wasp sting (alkaline): use vinegar (an acid) to neutralise the alkali. In each case the opposite substance neutralises the sting, so it hurts less. *A strong answer includes: the correct match + the word "neutralise".*
D5	Indigestion tablet	The tablet contains a base (e.g. magnesium hydroxide) that neutralises some of the extra stomach acid, making a salt and water. The pH rises a little, so the burning pain goes away.
D6	Lime on acidic soil	Lime (calcium hydroxide) is an alkali that neutralises the acid in the soil and raises the pH to a level where crops grow well. Far too much lime would make the soil alkaline, which is also bad for most crops.
D7	Toothpaste	Toothpaste is slightly alkaline, so it neutralises the acid made by bacteria. This stops the acid dissolving the tooth enamel, which helps prevent tooth decay.

E · Naming salts

Acid	Second part of the salt name
Hydrochloric acid	chloride
Sulfuric acid	sulfate
Nitric acid	nitrate

#	Acid	Alkali / base	Salt	Other product
E1	hydrochloric acid	sodium hydroxide	**sodium chloride**	water
E2	sulfuric acid	potassium hydroxide	**potassium sulfate**	water
E3	nitric acid	sodium hydroxide	**sodium nitrate**	water
E4	hydrochloric acid	calcium hydroxide	**calcium chloride**	water
E5	sulfuric acid	magnesium hydroxide	**magnesium sulfate**	water
E6	**hydrochloric acid**	potassium hydroxide	potassium chloride	water
E7	nitric acid	**calcium hydroxide**	calcium nitrate	water

#	Question	Model answer
E8	Table salt	Sodium chloride (NaCl). It can be made from hydrochloric acid + sodium hydroxide.

#	Question	Model answer
E9	Potassium nitrate fertiliser	nitric acid + potassium hydroxide → potassium nitrate + water. "Potassium" comes from the alkali; "nitrate" comes from nitric acid. *A strong answer includes: both reactants named + where each part of the name comes from.*

▶️ Check in the simulation

Every salt in E1–E9 can be checked in the Salt naming simulation (Explore → choose the acid and the alkali → React!): **school.taneski.com/science/g07/01-salt-naming/**

Scan to open

Unfamiliar situation challenge

Situation 1 — The swimming pool (pH should be 7.2–7.8; today pH 8.4)

#	Question	Model answer
S1a	Acidic, neutral or alkaline?	Alkaline (weakly alkaline): pH 8.4 is above 7.
S1b	Add dilute acid or alkali? Why?	Add a dilute **acid**. The water is too alkaline, so an acid will neutralise some of the alkali and lower the pH back into the 7.2–7.8 range.
S1c	Salt from HCl + sodium hydroxide	Sodium chloride (+ water).
S1d	Why add slowly and test again?	If too much acid is added at once, the pH could drop below 7.2 and the water would become too acidic, which also irritates eyes and skin and can damage the pool. Adding slowly and testing lets the manager stop when the pH is just right. *A strong answer includes: the risk of overshooting + the need to check the pH.*

Situation 2 — The mystery liquids (P red, Q green, R purple, S yellow)

#	Question	Model answer
S2a	pH of each liquid	P (red): about pH 0–2, strong acid. Q (green): pH 7, neutral. R (purple): about pH 12–14, strong alkali. S (yellow): about pH 5–6, weak acid.
S2b	Which could be water? Lemon juice?	**Q** could be pure water, because green means pH 7 (neutral). Lemon juice is most likely **P**: red means pH 0–2, and in C3 lemon juice has pH 2. Also accept **S** (yellow, weak acid) if the answer explains that lemon juice is a weak acid that is safe to drink. *A strong answer includes: a liquid for each + a reason based on the indicator colour and the pH.*
S2c	Mix P and R: final colour?	Green. P is an acid and R is an alkali; they neutralise each other, making a salt and water, so the pH becomes 7 (neutral).

#	Question	Model answer
S2d	Most dangerous to spill?	P or R. Both are at the extreme ends of the pH scale (strong acid pH 0–2, strong alkali pH 12–14), so they are the most corrosive and could burn skin and eyes. Q (neutral) and S (weak acid) are much less harmful. *A strong answer includes: a clear choice + a reason based on pH and "corrosive". Either P or R is fine if it is justified.*

Add your own situation and self-reflection

These sections have no fixed answers. When you check a partner's situation, look for: a realistic everyday context, a clear question, the correct topic (A–E), and an answer that uses correct chemistry words (acid, alkali, pH, neutralise, salt, mixture …). In your self-reflection, name specific topics and make a concrete revision plan.